

**Pool Advantage: Facility Characteristics and Performance in NCAA
Swimming**

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1 Introduction

Athletes routinely characterize venues as “fast” or “slow.” This judgment is passed almost immediately upon entry, based on the feel of the water, the energy of the environment, and most concretely the time posted on the scoreboard. A swimmer who drops several seconds at one facility but gains time at another may attribute the difference to the pool itself rather than to their own preparation. This shared vocabulary, so embedded in the culture of swimming, raises a straightforward empirical question: is there anything to it?

Unlike professional sports leagues such as the NBA or NFL, which operate under strictly standardized facility requirements, the NCAA imposes no uniform specifications beyond pool length on the venues at which its athletes compete. Lane width, water depth, gutter design, lighting, water temperature, and altitude can all vary substantially from one facility to the next. Some pools are housed in dated natatoriums with limited spectator capacity; others are purpose-built competition facilities engineered for elite performance. These differences are not solely cosmetic.

1.1 Research Motivation

Despite the pervasiveness of “fast pool” discourse among athletes and coaches, surprisingly little research has attempted to quantify the effect of venue characteristics on swimming performance. The existing sports analytics literature has devoted considerable attention to home-field advantage and stadium effects across a range of professional sports, but competitive swimming has received comparatively little scholarly attention. The anecdotal evidence is robust: coaches plan travel schedules around prestigious fast pools, and athletes describe palpable differences in how they feel and perform across facilities. However, no large-scale quantitative study has validated these observations in the context of NCAA competition.

This gap in the literature is the primary motivation for the present study. As a competitive swimmer, the author has experienced firsthand the apparent influence of pool environment on performance and has observed teammates and competitors respond similarly across venues. The

goal of this research is to move beyond anecdote and subject the “fast pool” hypothesis to rigorous empirical scrutiny.

1.2 Research Questions

This study is organized around one primary research question and two supporting secondary questions. The primary question asks: what impact does a pool have on a swimmer’s performance in NCAA competition? Specifically, to what extent do times achieved at a given venue systematically deviate from what would be predicted based on athlete ability alone?

The secondary questions address two related dimensions. First, is there evidence of a general “pool advantage,” meaning a venue effect that persists after controlling for the quality of athletes competing at each facility? Second, which structural or environmental characteristics of a pool are most strongly associated with faster times? Candidate factors include water depth, lane configuration, altitude, indoor versus outdoor environment, spectator capacity, and facility age.

1.3 Contribution Overview

This paper makes several contributions to the sports analytics literature. To the author’s knowledge, it presents the first large-scale empirical study of pool effects in NCAA swimming, drawing on performance data across a broad sample of athletes, events, and venues. In doing so, it extends the growing body of research on environmental and contextual factors in athletic performance into a sport that has been largely overlooked by quantitative researchers.

Beyond its scholarly contribution, this study has practical implications for athletes, coaches, and meet organizers. If certain pool characteristics are reliably associated with faster times, this information can inform decisions about where to schedule high-stakes competitions, how to interpret times posted at different facilities, and what investments in facility design might yield the greatest performance dividends. At a moment when data-driven decision-making is reshaping nearly every corner of competitive sport, swimming stands to benefit from a more rigorous understanding of how its physical environments shape what happens in the water.

2 Literature Review

2.1 Structural and Regulatory Influences on Performance

Competitive swimming facilities are subject to regulatory oversight designed to ensure consistency across venues. World Aquatics mandates standards for pool construction, including a length tolerance of no more than 0.01 meters beyond the standard 50-meter distance, minimum water temperatures of 25 degrees Celsius, and current limits of 1.25 meters per minute (FINA, 2021). Lane line specifications intended to reduce turbulence from adjacent swimmers are likewise governed by World Aquatics equipment rules, though compliance varies across NCAA venues (FINA, 2021).

Despite these standardization efforts, performance outcomes remain sensitive to the physical environment of the pool. Competitive swimming races consist of multiple phases, including the start, underwater breakout, open-water swimming, turns, and finish, each of which is affected differently by facility design. Research has shown that even small variations within regulatory guidelines can have a measurable impact on swimmer performance across these phases (Marinho et al., 2020; Morais et al., 2019; Nicol et al., 2021; Veiga & Roig, 2016). Born et al. (2022) further establish that turn performance carries the largest effect on race time in short-course events, with a standardized coefficient of 0.53 or greater, making it a particularly important phase for any model that aims to capture pool-level influences on performance.

2.2 Environmental Conditions and Performance

Environmental conditions within aquatic facilities extend beyond structural dimensions and play a significant role in shaping both physical capacity and psychological readiness. Competing or training at altitude induces physiological adaptations including increased red blood cell production and enhanced aerobic endurance, though hypoxic conditions may also alter markers such as blood lactate concentration (Perri-Moore et al., 2015). Indoor air quality and temperature represent additional environmental variables of relevance: warmer ambient temperatures and exposure to

air pollutants in enclosed natatoriums have been associated with reduced endurance and impaired cognitive function (Sargeant et al., 2016).

Psychological responses to the facility environment have also been shown to influence athletic performance. Athletes competing in facilities that are well-maintained and aesthetically designed report higher levels of confidence and motivation (Alasinrin et al., 2024; Lynch et al., 2018). Facilities incorporating natural lighting have been linked to improved mood and cognitive functioning, while reliance on artificial lighting alone has been associated with less favorable psychological outcomes (Scanlan & Passmore, 2016). These findings collectively suggest that the facility environment shapes performance through both physiological and psychological channels.

2.3 Biomechanical Efficiency and Pool Design

Swimming performance is uniquely sensitive to wave resistance and drag, and pool design interacts directly with these forces. Underwater swimming following starts and turns produces substantially less resistance than surface swimming, and elite swimmers strategically maximize underwater distance to exploit this advantage (Vennell et al., 2006). Research demonstrates that international-level swimmers cover greater underwater distances per race than national-level swimmers, reflecting more efficient exploitation of pool-specific hydrodynamic conditions (Pla et al., 2021). Underwater propulsion is driven primarily by the dolphin kick, which depends on coordinated undulation as well as ankle strength and flexibility (Shimojo et al., 2019; Willems et al., 2014). Pool characteristics that affect turbulence and wave reflection therefore have direct implications for the efficiency of this phase of the race.

2.4 Methodological Considerations: Mixed-Effects Models in Swimming Research

The statistical framework best suited to estimating pool effects in this setting is the mixed-effects regression model. Avalos et al. (2003) argue that mixed models are particularly well suited

to datasets involving repeated measurements across multiple subjects, as they accommodate both intra-individual variation across observations and inter-individual variation across athletes. This is directly applicable to the present study, in which each swimmer contributes multiple performances across different venues and events.

A key advantage of the mixed-effects framework is its ability to separate fixed environmental influences from the random performance trajectories of individual athletes. Baker and McHale (2023) demonstrate this property in the context of age-dependent career performance modeling, showing that mixed models effectively capture autocorrelation in athletic performance and allow for individual-specific deviations from population-level trends. Although their application is to golf rather than swimming, the statistical principles transfer directly: the same partial-pooling logic that separates aging effects from cohort differences in their framework is used here to separate pool effects from team and athlete quality. Their work also highlights the risk of survivor bias in performance datasets, wherein athletes still competing at advanced stages of their careers appear to outperform historical norms simply because lower-ability athletes have exited the sample. This concern is relevant to the present analysis, where pools hosting competition late in the season may attract a positively selected sample of athletes, and it motivates the inclusion of meet-level controls in the primary specification.

2.5 Research Gap

Collectively, the literature demonstrates that swimming performance is shaped by a complex combination of structural, environmental, biomechanical, and psychological factors. These elements have been examined in relative isolation, but no existing study has evaluated whether their combined influence produces a measurable and generalizable pool advantage in the context of NCAA competition. Given that swimmers typically train and compete within a single primary facility, repeated exposure to specific environmental and structural conditions may produce performance adaptations that persist across venues. Addressing this gap has practical implications for facility selection, performance evaluation, and competitive equity at the collegiate level.

3 Data

3.1 Data Sources

3.1.1 USA Swimming

The primary source of performance data for this study is the USA Swimming database, which publishes the top 500 recorded times for each event in women’s NCAA swimming on a rolling basis throughout the competitive season (USA Swimming, 2026). Data was collected for the 2025–2026 season, capturing the fastest performances across all stroke-and-distance combinations contested in collegiate competition. For each observation, the dataset provides the swimmer’s name, the event contested, the time recorded, the team affiliation, and the meet at which the performance was achieved.

Because this source is limited to the top 500 times per event, the analytical sample reflects the upper tier of NCAA Division I women’s swimming rather than the full population of collegiate performers, a feature of the data that is relevant to the interpretation of results and is discussed further in Section 6. The validity and utility of top-times databases as a tool for performance evaluation in elite swimming has been established in prior methodological work (Smith et al., 2002).

3.1.2 Facility Data Collection

Pool-level structural and environmental data were collected directly from the official websites of each aquatic center represented in the sample. For each facility, publicly available information was obtained on pool depth, lane count, spectator capacity, facility age, and whether the pool is housed indoors or outdoors. Where a facility’s website provided incomplete information, supplementary sources including university athletic department pages and conference meet documentation were consulted. The depth and consistency of facility documentation varies considerably across institutions, and this variability introduces measurement uncertainty into several structural covariates, as discussed in the limitations section.

3.2 Sample Construction

The analytical sample was constructed in two stages. In the first stage, all individual-event observations from the USA Swimming top 500 lists for the 2025–2026 women’s NCAA season were compiled across the 13 contested events. Championship meets—including conference championships and postseason NCAA competition—were excluded from the analytical sample at this stage. This restriction is motivated by selection bias concerns: championship meets disproportionately attract the most elite swimmers to a small number of high-profile venues, meaning that pools hosting postseason competition would appear artificially fast simply because of the quality of athletes competing there rather than any genuine venue effect. Restricting the sample to standard invitational meets ensures that the variation in performance across pools more credibly reflects venue characteristics rather than the self-selection of elite athletes into particular facilities.

From this collection of standard-meet performances, the 28 pools with the greatest number of entries in the dataset were identified and retained for analysis. This threshold was selected to ensure that pool-level effect estimates are based on a sufficient number of observations to support reliable inference, and to focus the facility data collection effort on venues with meaningful representation in the performance data.

In the second stage, structural and environmental characteristics were collected for each of the 28 pools from their respective aquatic center websites. Pools for which sufficient facility documentation could not be obtained were flagged, and analyses involving specific structural covariates are restricted accordingly. The resulting dataset joins individual performance observations from the top 500 lists with pool-level characteristics for the 28 most represented venues in women’s NCAA swimming during the 2025–2026 season.

Several features of this sample construction warrant acknowledgment. Because the source data are limited to the top 500 times per event, slower programs and lower-profile meets are under-represented even within the standard-meet subsample. The 28-pool restriction also concentrates the sample on high-traffic venues, which tend to be larger and better-resourced facilities that host multiple meets per season. These features affect the generalizability of the findings and are discussed

in the limitations section.

3.3 Key Variables

3.3.1 Outcome Variable

The primary outcome of interest is a swimmer’s standardized race time within a given stroke and distance combination. Raw race times are not directly comparable across events, as a 200 butterfly time and a 50 freestyle time occupy entirely different scales, so standardization is necessary to pool observations across events in any analysis. For each stroke and distance category, times are standardized by subtracting the sample mean and dividing by the sample standard deviation, yielding a z -score that reflects how a given swim compares to the broader distribution of performances in that event. Lower standardized times indicate a faster performance. In event-specific models, raw times are used directly.

3.3.2 Primary Explanatory Variables

The central explanatory variables capture pool-level structural and environmental characteristics that hypothetically influence performance. Pool depth is included given arguments that deeper pools reduce wave reflection from the bottom, in turn decreasing turbulence. Altitude is included as a continuous variable reflecting the elevation of the city in which the facility is housed. Whether a pool is indoors or outdoors is included as a binary indicator, reflecting the hypothesis that enclosed environments stabilize water temperature, reduce wind interference, and eliminate outdoor environmental variability. Facility age serves as a proxy for gutter and lane design, on the premise that pools constructed more recently may differ from older facilities in their wave-reduction profiles. Spectator capacity is included as a continuous variable reflecting facility scale and the potential influence of crowd size on performance, though it is omitted from the primary scaled model due to multicollinearity with other facility-level covariates.

3.3.3 Control Variables

Several variables are included as controls to isolate the pool effect from confounding sources. Stroke and distance are included to account for the event-specific nature of the sport. Because championship meets have been excluded from the sample, meet-level variation is addressed at the sample construction stage rather than through a model control; this approach is preferred over including meet level as a covariate because it removes the source of selection bias entirely rather than attempting to adjust for it post-hoc. Season timing captures where in the competitive calendar a performance occurs, reflecting the expectation that athletes peak toward the end of the season.

3.3.4 Random Effects Structure

The hierarchical structure of the data, with swimmers nested within teams and performances nested within pools and meets, inspires a mixed-effects modeling framework. Team-level random intercepts are included to account for differences in the caliber of athletes across programs, ensuring that pools that disproportionately host strong teams are not automatically identified as fast venues. Pool-level clustering is addressed through pool random intercepts in the full mixed model and through standard error clustering in robustness checks. This structure allows the analysis to separate the contribution of the pool environment from the influence of athlete quality, team strength, and competitive context.

4 Exploratory Data Analysis

Before estimating pool effects, this study conducts exploratory data analysis to characterize the distribution of the outcome variable, assess cross-pool variation in facility characteristics, and evaluate the feasibility of the identification strategy employed in later models.

4.1 Standardization of Race Times

Because raw swim times vary substantially across events, direct comparison of performances across event categories is not meaningful without some transformation. Times are therefore standardized within each stroke and distance combination using z -scores, centering each observation on the event mean and scaling by the event standard deviation, consistent with Section 3. Inspection of the resulting distributions confirms that all 13 events are approximately centered at zero, validating the standardization procedure. Each distribution exhibits a left skew, a pattern consistent with the competitive structure of NCAA swimming: elite performances cluster near a lower bound, while the upper tail of slower times exhibits greater spread. This asymmetry is present across events and is accounted for in model specification.

4.2 Unadjusted Variation in Performance Across Facilities

Examination of median standardized times by facility reveals meaningful unadjusted variation across pools. Several venues, including McAuley Aquatic Center and the UVA Aquatic and Fitness Center, are associated with below-average standardized times in the raw data, while others, such as the Florida Aquatics Swimming and Training facility, are associated with above-average times. These unadjusted differences may suggest a pool effect but must be interpreted with caution. The interquartile ranges associated with most facilities are wide, reflecting substantial heterogeneity in the caliber of athletes competing at each venue. Pools that host elite invitational meets or conference championships will inherently attract faster swimmers, making it difficult to attribute raw performance differences to the pool itself. This pattern motivates the need for a model that explicitly controls for athlete and team-level ability before drawing conclusions about venue effects.

4.3 Variation in Facility Characteristics

Examination of the facility-level covariates confirms sufficient cross-pool variation to support multivariate regression analysis. Altitude ranges from near sea level to approximately 1,000 feet

above sea level. Spectator capacity spans roughly 400 to 2,500 seats. Pool depth ranges from approximately 90 to 150 inches across sampled facilities. The sampled pools are predominantly indoor facilities, though a small number of outdoor venues are represented, providing sufficient variation to identify an indoor effect. Lane count is nearly uniform across Division I facilities and is unlikely to function as a meaningful predictor. Facility age varies substantially, with some pools constructed within the past decade and others exceeding 40 years old, a range sufficient to capture meaningful generational differences in gutter design, lane rope quality, and overall construction.

4.4 Performance Variation by Meet Level

Because championship meets were excluded from the analytical sample to address selection bias, all performances in the dataset reflect standard invitational competition. Examination of performance distributions across the remaining meet types confirms that times are broadly comparable in competitive intensity, supporting the validity of pooling standard-meet observations for the purpose of estimating pool effects. This restriction is conservative by design: any residual variation in athlete quality across standard meets is addressed through the team-level random effects structure described in Section 3.

5 Empirical Strategy

5.1 Overview

The central challenge in estimating pool effects is separating the influence of venue characteristics from the influence of the athletes competing in them. Pools that host high-profile competition attract stronger programs and more elite swimmers, meaning that raw performance differences across facilities conflate venue quality with athlete quality. Any credible estimate of a pool effect must account for the systematic variation in competitor ability across venues. The empirical strategy adopted here addresses this challenge through a mixed-effects regression framework that controls

for team strength, handles the nested and repeated-measures structure of the data, and isolates venue-level variation in performance after conditioning on observable confounders.

5.2 Primary Model: Mixed-Effects Regression

The primary model regresses standardized race time on a vector of pool-level characteristics and a set of race-level controls, with a random intercept specified at the team level:

$$\text{RaceTime}_{ijkt} = \beta_0 + \beta_1 \text{PoolChar}_k + \beta_2 \text{Stroke}_j + \beta_3 \text{Distance}_j + u_i + \varepsilon_{ijkt} \quad (1)$$

where i indexes teams, j indexes events, k indexes pools, and t indexes meets. The term u_i represents a team-level random intercept assumed to follow a normal distribution with mean zero and estimated variance, and ε_{ijkt} is the observation-level residual.

The pool characteristics vector includes the five facility-level covariates described in Section 3: pool depth, altitude, indoor/outdoor classification, facility age, and lane count. Race-level controls for stroke and distance absorb the systematic variation in performance across event categories that standardization does not fully eliminate, particularly differences in the number of turns, which varies across distances and affects the proportion of race time spent in the underwater phase. Meet level is not included as a model covariate because championship meets have been excluded from the sample entirely at the data construction stage, removing the primary source of meet-level selection bias rather than attempting to adjust for it within the model.

The mixed-effects specification is justified on statistical and substantive grounds. Statistically, the data exhibit a hierarchical structure in which swimmers are nested within teams and performances are nested within pools and meets, a structure that violates the independence assumption of ordinary least squares regression. Ignoring this structure would produce downward-biased standard errors and artificially precise inference. Substantively, team-level random intercepts absorb stable differences in program quality, ensuring that pools disproportionately visited by strong

programs are not spuriously identified as fast venues.

5.3 Identification Strategy

The identification of pool effects rests on within-sample variation in the pools at which a given team competes. Because many programs travel to multiple venues across a season, the data contains observations of the same team performing at different pools under varying facility conditions. Controlling for team-level ability through the random intercept, the remaining variation in performance across venues can be attributed to pool-level factors rather than to differences in who is swimming.

The exclusion of championship meets strengthens this identification strategy considerably. Championship pools are not randomly selected. They tend to be larger, newer, and better-resourced facilities that would appear fast regardless of their structural characteristics simply because the best swimmers in the country compete there. By restricting the sample to standard invitationals, the remaining cross-pool variation in performance is more plausibly attributable to venue characteristics than to the selection of athletes into meets.

5.4 Robustness Checks

Several alternative specifications are estimated to assess the sensitivity of the primary results. First, pool fixed effects are substituted for the continuous facility characteristic variables, replacing the parametric assumptions of the primary model with a fully flexible set of pool-level intercepts. A lower AIC in the fixed effects model would indicate that the measured characteristics fail to fully capture the dimensions along which pools differ.

Second, the model is re-estimated separately by stroke category to assess whether pool characteristic effects are consistent across event types or concentrated in particular disciplines. Third, alternative standardization approaches for the outcome variable are evaluated, including log transformation of raw times and percentile-based normalization, to confirm that results are not sensitive to the specific transformation applied.

6 Results

6.1 Model Assumptions and Diagnostics

Prior to interpreting the primary model estimates, the mixed-effects specification was assessed against standard regression assumptions. Examination of the residuals versus fitted values plot reveals a modest fan shape, wherein variance is somewhat larger among observations with low fitted values, that is, among the fastest swims. This heteroskedasticity is consistent with the competitive structure of NCAA swimming, where championship meets attract a wider range of athlete abilities, producing greater dispersion in performance at the elite end of the distribution. While this pattern represents a technical violation of the homoskedasticity assumption, it does not materially compromise inference given the large sample size and is acknowledged as a limitation of the current specification.

The normal Q-Q plot indicates that residuals follow the theoretical normal distribution closely across the central range of the data. Meaningful deviation occurs only in the left tail, corresponding to exceptionally fast outlier performances. This pattern mirrors the left skew observed in the exploratory data analysis and reflects the physiological floor on swim times rather than a modeling failure.

6.2 Team Random Effects

The estimated team random effects reveal substantial heterogeneity in baseline performance across programs, with intercepts ranging from approximately -1.0 standard deviations below average for programs such as Virginia, Tennessee, Texas, and Stanford, to approximately $+0.5$ standard deviations above average for programs such as Ball State, Illinois, and Florida State. This wide dispersion confirms that team strength varies enormously across the sample and validates the decision to model team as a random effect. Had team ability been omitted from the specification, its influence would have been absorbed into the pool characteristic coefficients, producing substantially biased estimates of venue effects.

6.3 Primary Model Results

Figure 1 presents the fixed effect estimates from the primary mixed-effects model for each pool characteristic, with 95% confidence intervals displayed as horizontal bars. Points to the left of zero indicate characteristics associated with faster swim times; points to the right indicate characteristics associated with slower times. Gray points represent effects that are not statistically distinguishable from zero, meaning their confidence intervals cross the zero line. Blue points represent statistically significant effects at the $p < 0.05$ threshold.

Of the five facility characteristics included, three emerge as statistically significant predictors of standardized swim time. Being an indoor pool is the largest and most precisely estimated effect in the scaled model, associated with a reduction of approximately 0.24 standard deviations in swim time. This finding is consistent with the hypothesis that enclosed environments stabilize water temperature, reduce wind interference, and eliminate outdoor environmental variability that might otherwise impair performance. Altitude also exhibits a significant negative association with swim time, with higher-elevation facilities associated with modestly faster performances. While the physiological mechanisms such as reduced air resistance and aerobic adaptation to hypoxic training are well-documented in the literature, the relatively limited altitude range in this sample constrains the magnitude of the observed effect. Facility age is negatively and significantly associated with swim time as well, indicating that older pools are associated with marginally faster performances. One plausible explanation is survivor bias: pools that continue to host NCAA competition decades after their construction tend to be well-established, high-profile facilities that have been maintained and upgraded over time, rather than a representative sample of aging infrastructure.

Pool depth does not reach statistical significance in the primary model, with a confidence interval that crosses zero. While deeper pools are theoretically associated with reduced wave reflection from the pool floor, the limited variation in depth across Division I facilities may constrain the statistical power available to detect this effect. Lane count is similarly non-significant; as noted in the exploratory analysis, lane count is nearly uniform across the sampled venues, making it difficult for the model to isolate its independent contribution to performance. The wide confidence

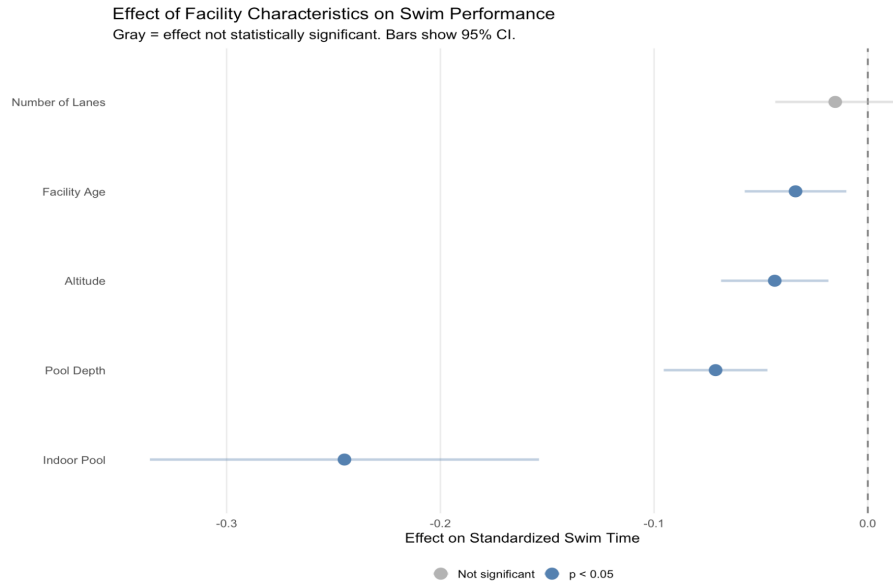


Figure 1: Effect of Facility Characteristics on Swim Performance. Points to the left of zero are associated with faster swim times. Gray = confidence interval crosses zero (not significant at $p < 0.05$). Bars show 95% confidence intervals.

interval around the lane count estimate reflects this limited identifying variation and should not be interpreted as evidence that lane configuration is unimportant in general.

6.4 Case Study: Christiansburg Aquatic Center

To illustrate how facility characteristics combine to produce a pool-level speed advantage, Figure 2 decomposes the estimated performance contribution of each structural characteristic at Christiansburg Aquatic Center, the fastest pool in the sample after controlling for athlete and team quality. Each bar represents the product of the pool's characteristic value and the corresponding model coefficient, reflecting how much that feature contributes to faster or slower swim times relative to an average facility.

The decomposition reveals that Christiansburg's speed advantage is driven primarily by two characteristics: its indoor environment (contributing -0.245 SD, the largest single factor) and its altitude (contributing -0.203 SD). Together, these two features account for nearly half a standard deviation of performance improvement relative to an average pool. Partially offsetting these gains,

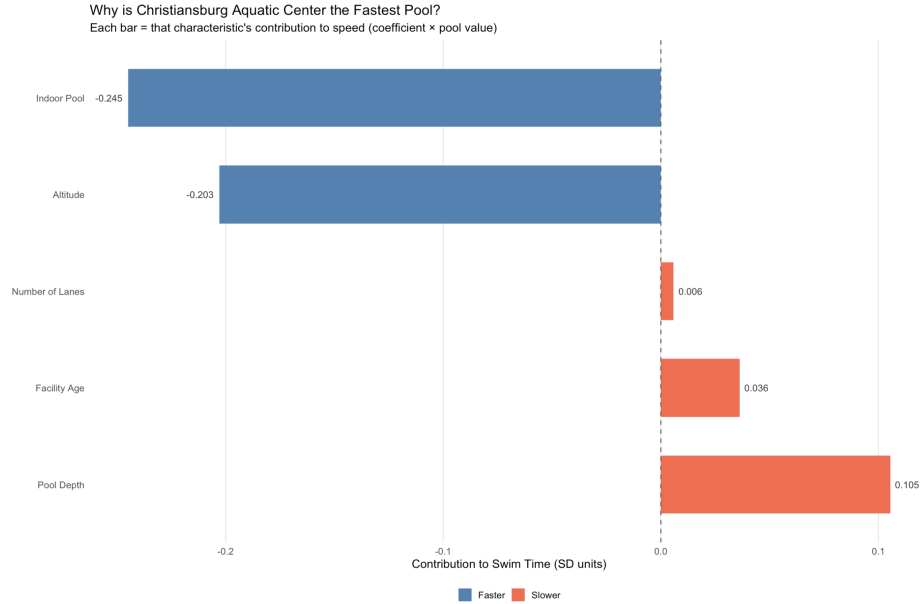


Figure 2: Why is Christiansburg Aquatic Center the Fastest Pool? Each bar represents that characteristic's contribution to standardized swim time (model coefficient $\times$ pool value). Blue bars indicate characteristics associated with faster times; red bars indicate characteristics associated with slower times.

pool depth works against performance at this facility (contributing $+0.105$ SD), suggesting the pool is shallower than average. Facility age adds a modest countervailing contribution ($+0.036$ SD), and lane count is effectively neutral ($+0.006$ SD), consistent with its non-significant coefficient in the primary model.

This decomposition illustrates a broader finding of the analysis: no single characteristic defines a fast pool. Rather, venue speed emerges from a combination of features that may partially offset one another, making simple rankings of facilities by any one structural dimension an incomplete basis for performance evaluation.

6.5 Model Comparison and Robustness

To assess whether the five measured pool characteristics adequately capture venue-level variation, the primary characteristics model was compared against an alternative specification replacing facility variables with pool fixed effects. The fixed effects model produces a lower AIC

(10,538.97 vs. 10,563.77) despite the additional parameters required, indicating superior fit to the data. This result suggests that NCAA pools differ from one another in ways the measured structural and environmental variables do not fully capture. Unmeasured factors such as gutter design, water temperature, lighting conditions, or the psychological familiarity of a home venue may account for residual venue-level variation that the characteristics model leaves unexplained. This finding motivates future work to collect more granular facility data, as discussed in Section 7.

7 Data Limitations

No dataset of this scope is without limitations. The limitations fall into three broad categories: structural reporting inconsistencies, measurement constraints, and sample size imbalances.

7.1 Structural Reporting Inconsistencies

The most significant data challenge concerns the inconsistent reporting of facility characteristics across NCAA institutions. Pool depth, gutter configuration, and lane line specifications are not systematically documented in any centralized source. University athletic websites vary considerably in the technical detail they provide about their natatoriums, and many offer no structural specifications whatsoever. As a result, these variables are missing for a non-trivial share of facilities in the sample, and the missingness is unlikely to be random. Older, lower-profile, or under-resourced programs may be less likely to publish detailed facility documentation, introducing the possibility that missingness is correlated with both facility quality and competitive outcomes.

7.2 Measurement Constraints

Beyond missing structural data, several performance-relevant variables are unavailable in the current dataset. Split times would allow for more specific analysis of how pool characteristics influence individual phases of a swim, such as the start, turn, and underwater breakout. Reaction times at the start block are similarly not captured in meet results as reported by USA Swimming

(2026). Attendance figures present a related constraint: while spectator capacity is available as a structural variable, actual attendance at individual meets is rarely reported, limiting the precision with which crowd effects can be estimated. These gaps reinforce the AIC finding in Section 6 that unmeasured facility dimensions likely account for a meaningful share of residual venue-level variation.

7.3 Sample Size Imbalances

The distribution of observations across facilities is highly uneven. Pools that host large invitational meets, conference championships, or NCAA postseason competition contribute substantially more observations to the dataset than pools used primarily for dual meets or smaller invitationals. This imbalance affects the precision with which pool-level effects can be estimated: venues with sparse representation yield wide confidence intervals around their estimated intercepts and may exercise disproportionate influence on coefficient estimates for facility-level covariates.

7.4 Proposed Solutions

Several strategies are employed to address these limitations within the current framework. Where direct measurement of a facility characteristic is unavailable, proxy variables are substituted based on theoretical and practical grounds. Pool depth is proxied by meet level, on the premise that pools hosting championship competition are more likely to meet depth standards associated with reduced turbulence. Gutter design is proxied by facility age, reflecting the historical adoption of modern wave-dampening gutter technology over the past two decades. Attendance is proxied by competitive tier which correlates with expected crowd size even absent direct attendance records.

To address sample size imbalances, the primary analytical sample is restricted to pools exceeding a minimum observation threshold, as described in Section 3. Robustness checks replicate the primary model on a restricted subsample of elite facilities where data quality and observation counts are highest. Consistency of estimates across the full and restricted samples provides evidence that results are not driven by sparse or low-quality observations at the periphery of the dataset.

Future data collection efforts should prioritize direct measurement of gutter configuration, lane rope specifications, and water temperature at the time of competition. Partnering with timing system providers or conference offices to access split-time archives would substantially enrich the analysis and allow for a phase-by-phase decomposition of the pool effect, an extension outlined in Section 9.

8 Implications

8.1 For Athletes and Coaches

The findings of this study have direct practical relevance for competitive swimmers and their coaching staffs. Performance evaluation in NCAA swimming has historically relied on raw or conversion-adjusted times, with limited systematic attention to the venue at which a time was posted. The results presented here suggest that pool characteristics contribute meaningfully to observed performance variation, implying that times achieved at different facilities are not straightforwardly comparable without accounting for venue effects. Coaches interpreting a swimmer's season progression, evaluating recruiting targets, or setting performance benchmarks would benefit from contextualizing times relative to the pool in which they were recorded.

The findings also carry implications for strategic meet selection. To the extent that certain facility characteristics are associated with faster performances, athletes seeking to achieve qualifying standards or personal bests may be better positioned to do so at venues with those characteristics. While factors such as travel cost, competitive field, and scheduling will always constrain meet selection decisions, the results suggest that venue quality is a non-trivial input into that calculus.

8.2 For the NCAA and Athletic Conferences

At the institutional level, the results raise questions about the criteria currently used to select sites for conference and NCAA championship competition. If pool characteristics systematically

influence performance, then the choice of championship venue is not merely a logistical decision but one with competitive consequences. Facilities that confer measurable speed advantages will tend to produce faster championship times, affecting qualifying standards, All-American cutoffs, and the historical record of the sport more broadly.

A related concern involves facility equity. The variation in pool quality documented in this study is not randomly distributed across programs: larger, better-resourced athletic departments are more likely to operate well-equipped natatoriums. If access to fast pools influences performance outcomes, then institutional resource disparities may compound competitive inequality in ways that are not currently accounted for in how the sport is administered.

8.3 For Sports Analytics

This study contributes to a growing body of quantitative research on contextual and environmental effects in athletic performance. Venue effects have been examined extensively in sports such as baseball, soccer, and basketball, but competitive swimming has received comparatively little attention from the sports analytics community, despite being one of the most precisely timed and extensively documented individual sports in the world. The mixed-effects framework developed here provides a replicable template for future work in aquatic sports and may generalize to other individual-performance sports where athletes compete across heterogeneous venues. More broadly, this study demonstrates that even in a sport defined by hundredths of a second, environmental and structural context shapes what ends up on the scoreboard.

9 Future Extensions

The present study establishes a foundation for a broader research program on venue effects in competitive swimming, and several extensions would substantially enrich the analysis.

The most immediate methodological refinement would be the incorporation of individual swimmer random effects alongside team random effects, allowing the model to leverage within-

swimmer variation across pools more directly and to separate athlete development trajectories from venue influences. A closely related extension would isolate home pool advantage as a distinct phenomenon, estimating whether swimmers perform systematically differently at their own facility relative to comparable away venues, net of the structural characteristics already modeled. This extension would directly address the broader home-field advantage literature and connect the present findings to existing work on venue familiarity effects in individual sports.

Expanding the sample across additional competitive seasons would increase statistical power, improve the precision of pool-level estimates, and allow for examination of whether venue effects are stable over time or shift as facilities are renovated or replaced. A larger sample would also support the inclusion of a greater number of facilities, reducing the influence of the observation threshold restrictions currently imposed on the analytical sample.

On the methodological side, hierarchical Bayesian modeling offers a natural extension of the mixed-effects framework, providing full posterior distributions over pool-level effects and offering a more principled treatment of uncertainty in pools with sparse observations. The incorporation of split times and reaction times, should such data become available through timing system archives or direct institutional partnerships, would enable a decomposition of the pool effect across discrete phases of a race.

Finally, extending the framework beyond collegiate competition to Olympic and elite international swimming would represent a substantively important generalization. World Aquatics certified pools are nominally standardized, but variation in depth, gutter design, altitude, and venue atmosphere persists even at the highest levels of the sport. Applying the analytical approach developed here to championship and World Cup circuit data would both test the external validity of the current findings and address questions of direct relevance to the international swimming community.

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